

Feature Engineering

Terms and Concepts — A Glossary

Brief definitions of key terms from the feature-engineering curriculum, grouped alphabetically.

HOW TO USE THIS GLOSSARY

Definitions are written in the context of the feature-engineering course — that is, for building and evaluating machine-learning models on tabular data. Terms are grouped by first letter. Where a term names a scikit-learn class or parameter, the definition notes what it does in a pipeline.

A

Accuracy — The proportion of predictions a classification model gets right, out of all predictions made. A simple headline metric, but misleading on imbalanced data.

Adjusted R-squared — A version of R-squared that penalizes adding features that don't improve the model, so it doesn't automatically rise every time a variable is added.

Aggregation features — New features built by summarizing several raw values into one, such as a sum, mean, or count over a group of records.

Arithmetic feature combinations — New features formed by combining existing columns with arithmetic operations — adding, subtracting, multiplying, or dividing them.

Arrays — Ordered grids of numbers (typically NumPy arrays) that hold the numeric data machine-learning algorithms operate on.

Average call duration — A derived telecom feature: total call minutes divided by the number of calls, giving a per-call average.

Average service-call charges — A derived feature capturing the typical cost of a customer's service calls, often used in churn modeling.

B

Baseline regression model — A simple first model built before feature engineering, used as a reference point to measure whether later features actually help.

Bell-shaped curve — The symmetric, single-peaked shape of a normal (Gaussian) distribution, with most values near the mean and tails falling off on both sides.

Binary classification — A prediction task with exactly two possible output classes, such as churn / no-churn or spam / not-spam.

Binary variables — Features that take only two values, usually encoded as 0 and 1, such as an indicator flag.

Body mass index (BMI) — A domain-specific derived feature computed from weight and height — a classic example of feature creation using domain knowledge.

Box-Cox transformation — A power transformation that reshapes positive, skewed data toward normality by finding an optimal exponent. Requires strictly positive values.

C

Categorical columns — DataFrame columns whose values are labels or categories rather than numbers, needing encoding before modeling.

Categorical data — Data representing discrete groups or labels (colors, cities, plan types) rather than measured quantities.

Categorical features — Input features that hold category labels; they must be encoded into numbers for most algorithms.

Categorical feature names — The list of column names identified as categorical, used to route those columns to a categorical transformer in a pipeline.

Categorical transformer — The branch of a ColumnTransformer that preprocesses categorical columns, typically applying one-hot or other encoding.

Category grouping — Combining rare or related categories into broader buckets to reduce sparsity and noise before encoding.

Churn prediction — The task of predicting which customers will stop using a service — the running example in this feature-engineering material.

Classification model — A model that assigns inputs to discrete classes rather than predicting a continuous number.

ColumnTransformer — A scikit-learn tool that applies different preprocessing steps to different subsets of columns (e.g., scaling numerics, encoding categoricals) in one object.

Computational cost — The time and compute resources a transformation or model requires — a practical constraint in feature engineering, especially at scale.

Constant term — The intercept added to a regression, and the bias column PolynomialFeatures can insert (controlled by `include_bias`).

Correlation — A measure of how strongly two variables move together, ranging from -1 (opposite) through 0 (none) to +1 (identical direction).

Correlation analysis — Examining pairwise correlations among features and with the target to spot redundancy, multicollinearity, and predictive signal.

Correlation matrix — A table showing the correlation between every pair of features, often visualized as a heatmap.

Count encoding — Replacing each category with the number of times it appears in the data, turning a label into a numeric frequency count.

CSV data — Tabular data stored as comma-separated values — a common raw input format loaded into a DataFrame.

Cubic polynomial — A degree-3 polynomial feature expansion, adding squared and cubed terms to capture stronger curvature (higher overfitting risk).

Cumulative explained variance — The running total of variance explained as principal components are added, used to choose how many components to keep.

Curse of dimensionality — The problems that arise as features multiply — data becomes sparse, distances lose meaning, and models overfit — motivating dimensionality reduction.

D

Data cleaning — Fixing raw-data problems (missing values, wrong types, duplicates, outliers) so the data is fit for modeling.

DataFrame — The primary pandas table structure — labeled rows and columns — used throughout feature engineering.

Data leakage — When information unavailable at prediction time (often from the future or the target) sneaks into features, giving falsely optimistic results.

Data normality — How closely a variable's distribution matches a normal (Gaussian) shape — something transformations often aim to improve.

Data preprocessing — All steps that prepare raw data for modeling: cleaning, encoding, scaling, transforming, and reducing features.

Data representation — How information is encoded as features — a good representation makes patterns easier for a model to learn.

Data scaling — Rescaling numeric features to comparable ranges (e.g., standardization or min-max) so no feature dominates by magnitude alone.

Data transformation — Applying a function to feature values (log, power, scaling, encoding) to make them more suitable for modeling.

Dataset — The full collection of records and features used to train and evaluate a model.

Decision trees — Models that split data on feature thresholds in a branching structure; the building blocks of random forests.

Derived metrics — Features computed from other columns (ratios, totals, averages) rather than measured directly.

Dimensionality reduction — Reducing the number of features while keeping most information, either by selecting a subset or projecting into a lower-dimensional space (e.g., PCA).

Divide-by-zero errors — Errors from division when a denominator is zero — a hazard when engineering ratio features, requiring guards or safe transforms.

Domain expertise — Deep knowledge of the problem area, essential for knowing which engineered features will actually be predictive.

Domain knowledge — Understanding of the subject matter that guides feature creation and selection — repeatedly stressed as critical to feature engineering.

Domain-specific features — Features designed using subject knowledge (like BMI in health), tailored to the particular problem.

E

Elbow plot — A chart used to pick a parameter (like the number of PCA components or clusters) by locating the 'elbow' where added components stop helping much.

Encoding — Converting categorical labels into numeric form (one-hot, count, frequency, target encoding) so algorithms can use them.

Engineered features — New or transformed features created deliberately to improve model performance, as opposed to raw input columns.

Explanatory variables — The input variables (X) used to predict the target; also called predictors or independent variables.

Explained variance — The amount of total data variance captured by a component or model — in PCA, how much information each principal component retains.

Explained variance ratio — The fraction of total variance attributable to each principal component, guiding how many components to keep.

Extraneous features — Features that add no predictive value and may hurt the model by adding noise or dimensionality.

F

F1 score — The harmonic mean of precision and recall, giving a single balanced metric useful on imbalanced classification problems.

Feature creation — Generating new, meaningful features from existing data — one of the core activities of feature engineering.

Feature engineering — Thoughtfully selecting, transforming, and creating the X variables used by a model to boost accuracy and reliability; tied to the specific problem and target.

Feature importance — A measure of how much each feature contributes to a model's predictions, commonly reported by tree-based models.

Feature importance scores — The numeric values ranking features by their contribution, used to guide feature selection.

Feature names — The labels identifying each column, important for interpreting models and mapping transformed features back to their source.

Feature selection — Identifying and keeping the most useful features while discarding irrelevant or redundant ones to reduce overfitting and cost.

Fit quality — How well a model captures the underlying pattern in the data — neither underfitting nor overfitting.

fit method — The scikit-learn method that learns parameters from training data (e.g., a scaler's mean, PCA's components) without transforming it.

fit_transform method — A convenience method that learns parameters and applies the transformation in one step; used on training data, never on test data.

Frequency encoding — Replacing each category with how often it occurs, expressed as a proportion of the total (a normalized count).

FunctionTransformer — A scikit-learn wrapper that turns any custom function (like `log1p`) into a pipeline-compatible transformer.

G

Gamma squeeze — A finance-domain example term: a rapid price rise driven by options hedging — illustrating domain-specific derived signals.

Gaussian distribution — Another name for the normal distribution — the symmetric bell curve many statistical methods assume.

Gaussian noise — Random variation drawn from a normal distribution, sometimes added to data or present as measurement error.

Generalization — A model's ability to perform well on new, unseen data rather than just the data it was trained on.

H

handle_unknown='ignore' — A `OneHotEncoder` setting that lets the encoder ignore categories not seen during training instead of raising an error at inference.

Heatmap — A color-coded grid used to visualize a matrix of values, such as a correlation matrix.

Heteroscedasticity — When the spread of errors or a variable changes across its range; transformations like log can stabilize this uneven variance.

Hierarchical categories — Category values with nested levels (e.g., country > state > city) that may need special encoding or grouping.

High-dimensional feature sets — Datasets with many features, prone to the curse of dimensionality and good candidates for dimensionality reduction.

Histogram — A bar chart of a variable's value distribution, used to spot skew, outliers, and shape before transforming.

Homogeneity — The degree to which a group of values is similar; relevant when assessing categories or cluster quality.

House age feature — An engineered feature (e.g., current year minus year built) — an example of creating a meaningful variable from raw dates.

I

Imbalanced dataset — A dataset where classes are very unequally represented, which can bias accuracy and require careful metric choice.

Impurity — A measure (like Gini or entropy) of how mixed the classes are in a tree node; reductions in impurity drive tree-based feature importance.

Imputation — Filling in missing values with substitutes such as the mean, median, or a model-based estimate.

include_bias parameter — A PolynomialFeatures option controlling whether a constant (bias) column of 1s is added to the expanded features.

Inference — Using a trained model to make predictions on new data — the deployment-time counterpart to training.

Input features — The X variables fed into a model to produce predictions.

Interaction features — Features formed by multiplying two or more variables together to capture effects that depend on their combination.

Interaction terms — The cross-product terms (like $x_1 * x_2$) added by polynomial expansion to model how features jointly affect the target.

interaction_only parameter — A PolynomialFeatures option that includes only interaction cross-terms and omits pure power terms like x -squared.

Interpretability — How easily humans can understand why a model makes its predictions; often reduced by heavy feature engineering and PCA.

Irrelevant features — Features unrelated to the target that add noise and can degrade performance; candidates for removal.

K

Kaggle datasets — Publicly available datasets from the Kaggle platform, frequently used for practice and examples.

Kitchen-sink regression — Throwing every available feature into a model without selection — a naive approach prone to overfitting and poor interpretability.

L

Lag features — Features built from earlier time steps of a series (e.g., yesterday's value), used to give time-series models memory.

Latent factor analysis — Uncovering hidden underlying variables that drive the observed features; PCA is one way to surface such latent structure.

Linear fit — A model that assumes a straight-line relationship between features and target.

Linear relationship — A relationship where the target changes proportionally with a feature, well captured by linear models.

Linear regression — A model predicting a continuous target as a weighted linear combination of features — often the baseline model here.

LinearRegression — The scikit-learn class implementing ordinary least-squares linear regression.

List comprehension — A concise Python syntax for building lists in one line, often used to generate feature names or column lists.

Log transformation — Replacing values with their logarithm to compress large values, reduce right skew, and pull in long tails.

Long-tailed distribution — A distribution with a small number of extreme values stretching far from the bulk, often addressed with a log transform.

M

Machine learning (ML) — Building models that learn patterns from data to make predictions or decisions without being explicitly programmed for each case.

Mean — The arithmetic average of a set of values; the center used by standardization and mean imputation.

Mean absolute error (MAE) — A regression metric averaging the absolute differences between predictions and actual values; robust to outliers.

Mean imputation — Filling missing numeric values with the column's mean — simple but can distort variance and relationships.

Median — The middle value when data is sorted; a robust center used for imputation when data is skewed.

Min-max scaling — Rescaling features to a fixed range (usually 0 to 1) using the minimum and maximum values.

MinMaxScaler — The scikit-learn transformer that performs min-max scaling.

Missing values — Absent entries in the data that must be imputed or otherwise handled before modeling.

Model accuracy — How correct a model's predictions are, measured by an appropriate metric for the task.

Model complexity — How flexible a model is; higher complexity (e.g., high-degree polynomials) fits training data better but risks overfitting.

Model evaluation — Assessing model performance with metrics on held-out data to judge how well it generalizes.

Model fit — How well a model matches the data — the balance between underfitting and overfitting.

Model performance — The overall quality of a model's predictions across relevant metrics.

Model predictive power — The strength of a model's ability to forecast the target accurately on new data.

Model reliability — How consistently a model performs across different data and conditions.

Moving window — A sliding span over sequential data used to compute rolling statistics like moving averages.

Multistep pipeline — A pipeline chaining several transformations and a model so they run in sequence and fit only on training data.

N

Natural logarithm of 1 + X (log1p) — The $\log(1 + x)$ transform, which handles zeros safely and reduces right skew without breaking on $x = 0$.

Negative correlation — A relationship where one variable rises as the other falls (correlation below zero).

Negative infinity (-inf) — The value log produces at zero; a reason to use log1p or shift data before a plain log transform.

Noise — Random, non-informative variation in data that obscures the underlying signal.

Nonlinear relationships — Relationships that curve rather than follow a straight line, captured with polynomial features or other transforms.

Normal distribution — The symmetric bell-shaped distribution assumed by many methods; transformations often aim to approximate it.

Normalized frequency — A category's occurrence count expressed as a proportion of the total, as produced by frequency encoding.

Null values — Empty or missing entries in a dataset (NaN/None), requiring handling before modeling.

NumPy — The core Python library for fast numerical arrays and math, underpinning pandas and scikit-learn.

Numeric columns — DataFrame columns holding numbers, routed to a numeric transformer for scaling or transformation.

Numeric data — Measured quantities (integers or floats) that algorithms can process directly.

Numeric features — Input features that are numeric, as opposed to categorical.

Numeric transformer — The pipeline branch that preprocesses numeric columns, typically imputing and scaling them.

O

One-hot encoding — Turning a categorical variable into a set of 0/1 columns, one per category, without implying any order between categories.

OneHotEncoder — The scikit-learn transformer that performs one-hot encoding.

Orthogonal principal components — PCA components that are mutually perpendicular and therefore uncorrelated with one another by construction.

Out-of-sample inference — Making predictions on data not used during training — the true test of generalization.

Outliers — Extreme values far from the rest of the data that can distort scaling, fitting, and metrics.

Overfitting — When a model learns the training data too closely, including its noise, and fails to generalize to new data.

P

Pandas — The Python library providing DataFrames and tools for loading, cleaning, and manipulating tabular data.

pandas DataFrame — The labeled table object from pandas used to hold and transform feature data.

PCA components — The new orthogonal axes produced by PCA, each a linear combination of original features capturing a share of variance.

Permutation importance — A model-agnostic way to measure feature importance by shuffling one feature and seeing how much performance drops.

Pipeline — A scikit-learn object that chains preprocessing steps and a model so they apply consistently and avoid leakage.

Pipeline fitting — Calling fit on a pipeline, which fits each step in turn on the training data.

Pipeline steps — The named stages of a pipeline (e.g., scaler, encoder, model) executed in order.

Positive correlation — A relationship where two variables rise and fall together (correlation above zero).

Positive infinity (inf) — A value that can arise from divide-by-zero or overflow; must be handled when engineering features.

Power transformation — A family of transforms (Box-Cox, Yeo-Johnson) that reshape skewed data toward a normal distribution and stabilize variance.

PowerTransformer — The scikit-learn transformer applying Box-Cox or Yeo-Johnson power transformations.

Precision — Of the items a model labeled positive, the fraction that truly are positive.

predict method — The scikit-learn method that returns a fitted model's predictions for given inputs.

predict_proba method — The scikit-learn method returning predicted class probabilities rather than hard labels.

Prediction — A model's estimated output for a given input.

Predictive capability — How well features or a model can forecast the target.

Predictive features — Features that carry genuine signal about the target and improve predictions.

Preconfigured pipeline — A pipeline assembled in advance with its steps defined, ready to fit on data.

Preprocessing — The transformation stage that readies raw data for a model — cleaning, encoding, scaling, and reducing features.

Principal component analysis (PCA) — A dimensionality-reduction technique that projects correlated features onto a smaller set of uncorrelated components capturing most variance.

Principal components — The uncorrelated directions of maximum variance that PCA extracts from the data.

Principal-component count (n_components) — The PCA setting for how many components to keep, either a fixed number or a target share of variance.

Product feature — An engineered feature created by multiplying two columns together, a form of interaction feature.

Put-call ratio (PCR) — A finance-domain derived feature comparing put and call option volume — an example of domain-specific signal.

Q

Quadratic polynomial — A degree-2 polynomial expansion adding squared and interaction terms to capture gentle curvature.

R

Random error term — The unexplained, random component of a target that no feature can account for.

Random forest — An ensemble of many decision trees whose votes are combined, providing strong predictions and feature importance scores.

RandomForestClassifier — The scikit-learn class implementing a random-forest model for classification.

Random seed — A fixed starting value for random number generation that makes results reproducible across runs.

Raw data — Data in its original, unprocessed form before cleaning and feature engineering.

R-squared (R2 score) — A regression metric giving the fraction of target variance explained by the model, from 0 to 1 (can go negative for poor fits).

Recall — Of all the truly positive items, the fraction the model correctly identified.

Redundant features — Features that duplicate information already present in others, adding little and inflating dimensionality.

Regression analysis — Modeling a continuous target as a function of input features.

Regression metrics — Measures like MAE, RMSE, and R-squared used to evaluate models that predict continuous values.

Regression model — A model that predicts a continuous numeric target.

Regularization — Techniques that penalize model complexity to curb overfitting, encouraging simpler, more general models.

Renovation indicator feature — An engineered binary flag marking whether a property was renovated — an example of a domain-informed feature.

Reproducibility — The ability to obtain the same results by fixing random seeds and controlling the process.

Right-skewed distribution — A distribution with a long tail toward high values; commonly corrected with a log or power transform.

ROC area under the curve (ROC AUC) — A classification metric summarizing how well a model separates classes across all thresholds; 0.5 is random, 1.0 is perfect.

Rolling average — The mean computed over a moving window of sequential data, smoothing short-term fluctuations.

Rolling standard deviation — The standard deviation over a moving window, capturing how variability changes over time.

Root mean squared error (RMSE) — A regression metric — the square root of average squared error — that penalizes large errors more heavily and shares the target's units.

S

Scaling — Adjusting the range or distribution of numeric features so they are comparable and well-behaved for modeling.

Scatter plot — A plot of one variable against another to reveal relationships, trends, and outliers.

Scikit-learn — The primary Python machine-learning library, providing transformers, models, pipelines, and metrics.

Scikit-learn pipeline — A pipeline built with scikit-learn to chain preprocessing and modeling steps cleanly.

Scree plot — A plot of each principal component's explained variance, used to decide how many components to retain.

Signal — The genuine, informative pattern in data that a model aims to learn, as opposed to noise.

Skewed data — Data whose distribution is asymmetric, leaning toward one tail; often reshaped with transforms.

Skewness — A numeric measure of a distribution's asymmetry; positive means a long right tail, negative a long left tail.

sparse_output=False — A OneHotEncoder setting that returns a dense array instead of a sparse matrix, convenient for inspection and some workflows.

Square-root transformation — Replacing values with their square root to moderately reduce right skew — gentler than a log transform.

Squared terms — The x-squared features added by polynomial expansion to capture curvature.

Standard deviation — A measure of how spread out values are around the mean; the scale used in standardization.

Standard normal distribution — A normal distribution with mean 0 and standard deviation 1 — the target shape of standardization.

Standardization — Rescaling a feature to mean 0 and standard deviation 1, so features share a common scale.

Statistical aggregation — Summarizing multiple values into statistics like sums, means, or counts to form new features.

Sum feature — An engineered feature adding several columns together, such as total charges from monthly charges.

Symmetric distribution — A distribution that mirrors itself around its center, like the normal curve.

Synthetic features — Artificially constructed features derived from existing data to expose patterns to the model.

T

Target encoding — Replacing a category with a statistic of the target for that category (like its mean); powerful but prone to leakage if done carelessly.

Target leakage — A form of data leakage where a feature encodes information about the target that wouldn't be available at prediction time.

Target variable (y variable) — The outcome the model is trained to predict.

Telecom churn dataset — The customer-churn dataset used as the worked example throughout this material.

Test data — Held-out data used only to evaluate a trained model's generalization, never to fit transformers or the model.

Test set — The specific partition of data reserved for final evaluation.

Time-series data — Data indexed by time, where order matters and features like lags and rolling statistics apply.

Total-call feature — An engineered feature summing a customer's calls across categories into a single count.

Total-charge feature — An engineered feature summing charge components into a customer's overall charge.

Total-minute feature — An engineered feature summing call minutes across categories into a single total.

Training data — The data used to fit transformers and the model; all learned parameters come from here.

Training set — The specific partition of data used for fitting the model.

train_test_split — The scikit-learn function that splits data into training and test sets, done before fitting to prevent leakage.

Transform method — The scikit-learn method that applies a previously fitted transformation to data.

Transformation pipeline — A chain of data transformations applied in sequence, often wrapped in a scikit-learn Pipeline or ColumnTransformer.

Tree-based models — Models built from decision trees (including random forests) that handle nonlinearities and yield feature importances.

U

Uniform distribution — A distribution where every value in a range is equally likely — flat, with no peak.

Uncorrelated features — Features with no linear relationship to each other; PCA produces uncorrelated components by design.

Unseen categories — Category values that appear at inference but not in training, handled via settings like `handle_unknown='ignore'`.

V

Variance — A measure of how spread out values are; in PCA, the quantity each component seeks to maximize and preserve.

Variance stabilization — Transforming data so its variance is more constant across the range of values, a common goal of power transforms.

X

X variables — The input features used to predict the target; the focus of feature engineering.

Y

y variable — The target variable a model is trained to predict.

Yeo-Johnson transformation — A power transformation like Box-Cox but able to handle zero and negative values, reshaping data toward normality.